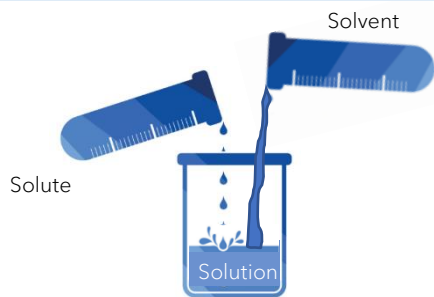




CONCENTRATION

Concentration is defined as the amount of solute present in a certain amount of solvent or solution. **Solute** is a substance that is dissolved in a solvent to create a solution. **Solvent** is a substance that dissolves a solute to create a solution.

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Types of Solutions Types of Aqueous Solutions

A **solution** is a homogenous mixture of two or more substances. Types of solutions according to how solutes are dissolved:

1. **Saturated** – a solution that has maximum amount of solute that are dissolved Excel Review Center
2. **Unsaturated** – a solution that has less amount of solute than the solvent can possibly dissolve
3. **Supersaturated** – a solution that has an amount of solute greater than the amount of solute in a saturated solution.

Solutes that are dissolved in water are called **aqueous solutions**. There are two types of aqueous solutions:

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1. **Electrolytes** – these are substances that can conduct electricity when they are dissolved in water. A good electrolyte must dissociate or must break into cations and anions once it is dissolved. Most acids and bases are also considered electrolytes, although some of them are weak electrolytes.
2. **Non-electrolytes** – these are substances that cannot conduct electricity when they are dissolved in water.

Different Types of Concentration Units

Percent by Mass

It is the mass of the solute in a specific mass of a solution. It is unitless. It is independent of variations in temperature since this involves mass.

$$\% \text{ by mass} = \frac{\text{mass solute}}{\text{mass solution}} \times 100$$

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Molarity

It is the number of moles of solute present in a liter of the solution. It has units of mol/L or M.

$$\text{Molarity} = \frac{\text{moles solute}}{\text{vol. solution in liters}}$$

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Molality

It is the number of moles of solute present in 1 kg of solvent. It has units of mol/kg or m. Like percent by mass, this is also independent on changes in temperature.

$$\text{molality} = \frac{\text{moles solute}}{\text{mass of solvent in kg}}$$

Normality

It is the molar concentration divided by an equivalent factor or it is a measure of concentration equal to the gram equivalent weight per liter of solution.

$$\text{normality} = \frac{\text{equivalent weights}}{\text{liters of solution}}$$

Acid

Any substance which can dissolve into the water produces hydrogen ions.

- ✓ Acids taste sour
- ✓ Acids changes color of litmus paper to red
- ✓ Acids react with some metals to produce hydrogen gas
- ✓ Acids react with carbohydrates and bicarbonates to produce carbon dioxide gas
- ✓ Aqueous acid solutions are electrically conductive

Concentration is $> 1.0 \times 10^{-7}$
pH < 7

Chemical reactions between acids and bases will always produce water and salt.

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Base

Any substance which can dissolve into the water produces hydroxide ions.

- ✓ Bases taste bitter
- ✓ Bases changes color of litmus paper to blue
- ✓ Bases feel slippery
- ✓ Aqueous base solutions are electrically conductive.

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Concentration is $< 1.0 \times 10^{-7}$
pH > 7

Electronegativity is the measure of the ability of atoms to attract electrons toward itself. The Pauline scale is the most commonly used. **Fluorine** is the most electronegative element with a value of 4.0 while **Francium** is the least electronegative with a value of 0.7

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The Tyndall Effect

Named after the English physicist, John Tyndall, this is one method of distinguishing a colloid from a solution. It states that when light passes through a colloid, it should be scattered by the dispersed particles. This scattering will not occur in solutions because the solute molecules are too small to affect the light passing through it.

Dilution is the process of making a concentrated solution into a less concentrated one. This process is done by adding more solvent to the given solution without changing the number of moles of the solute present.

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Moles of solute before dilution = Moles of solute after dilution

$$\text{Molality}_{\text{initial}} \text{Volume}_{\text{initial}} = \text{Molarity}_{\text{final}} \text{Volume}_{\text{final}}$$